RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105

RAJALAKSHMI ENGINEERING COLLEGE

SOFTWARE CONSTRUCTION

(CS23432)

Laboratory Observation Note Book

Name : SAHYUKTHA S.S

Year / Branch / Section : II nd | B. Tech. IT

Register No. : 2116241001217

Semester : IV th

Academic Year : 2025 - 2026

RAJALAKSHMI ENGINEERING COLLEGE (AUTONOMOUS)
RAJALAKSHMI NAGAR, THANDALAM – 602 105

BONAFIDE CERTIFICATE

NAME SAHMUKTHA . S.S REGISTER NO. 241001217

ACADEMIC YEAR 2025-26 **SEMESTER**- IV **BRANCH**: B. Tech. IT.

This Certification is the Bonafide record of work done by the above student

in the SOFTWARE CONSTRUCTION (CS23432) Laboratory

during the year 2025- 2026.

Signature of Faculty -in – Charge

Submitted for the Practical Examination held on 12.5.26

Internal Examiner External Examiner

Student management System

EXP NO: 01
DATE:

AZURE DEVOPS ENVIRONMENTS SETUP.

AIM:

To set up and access the Azure Devops enviroment by creating an organization through the Azure portal.

INSTALLATION:

1. open your Web browser and go to the Azure site https://azure.microsoft.com/en-us/get-started/azure-portal sign in using your Microsoft account credentials If you don't have a Microsoft account, you can create on here: https://azure//signup.live.com/2lis=1

get started with Azure sign in

2. Azure home page
Azure services - Azure Devops organization
3. open Devops environments in the Azure platform by typing Azure Devops organization in the search bar
4. click on the My Azure Devops organization link and create an organization Home page.
Azure Devops.
My Azure Devops Organizations

Result: successfully accessed the Azure Devops environment and created a new organization through the Azure portal.

EX NO: 2

AZURE DEVOPS PROJECT SETUP AND USER STORY MAN

AIM:

To setup an Azure Devops project for efficient collaboration and Agile work management

PROCEDURE:

1. create an Azure Account

Azure Devops
Samyuktha.SS-2024it@rajalakshmi.edu.in

2. create the first project in your organization
dev.azure.com/

(a) After the organization is setup, you'll need to create your first project. this is where you'll begin to manage code, pipelines, work items and more

(b) on the organization's Home page, click on the New project button.

(c) Enter the project name, description and visibility

Name: choose a name for the project
Description: optionally, add a description to provide more context about the project.
visibility: choose whether you want the project to be private or public

d) once you've filled out the details, click create to set up your first project.

Create new project

project name: Student Data Management System (edu track)

visibility: Public

version control: Git

Work item process:

3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations you can switch between them from the top left corner (next to your user profile). click on the organization name, and you should be taken to the Azure Devops Organization Home page.

projects:

4. <u>Project dashboard</u>

<u>About this project:</u>

The Student Data Management System (edu track) is a web-based application used to manage student records efficiently. it allows instructors to add, update delete, and view student details. The system supports bulk data import using CSV files with validation it also provides search functionality for quick access to student information.

5. <u>To Manage User stories</u>

User stories define features like adding students, uploading CSV files, and searching records. In Azure Devops, these are managed using Boards and work items. Each story represents a function of the system and can be assigned and tracked. This helps in organizing and completing tasks efficiently

Boards → Work items → + new Work items → User Story

Result:

Successfully created an Azure Devops project with user story management and Agile Workflow setup

EX NO 3 setting up Epics, Features and user stories for project Planning

AIM:

To learn about how to create epics, user story features, back logs for you assigned project

PROCEDURE:

Create epic, Features, user stories, Task.

1. Fill in epic

Epic 1.

Student Data Management and Batch Processing system.

2. Fill in Features:

- User Registration and Login
- Student Record Management (CRUD)
- CSV File Upload for Bulk Import
- Student Search Functionality
- Data Validation and Error Handling

3. Fill in user Story Details:

Student Record Management:

As an instructor, I want to add, update, and delete student details so that I can manage student Records efficiently.

CSV Upload system:

As an instructor, I want to uploadcsv files so that I can import multiple student records at once.

Student Search System:

As an instructor, I want to search student details so that I can quickly find required information.

Result: Thus, the creation of epics, features, user stories, and tasks has been completed successfully.

Ex No: 4

Sprint Planning

AIM:

To assign user stories to specific sprints for the Student Data Management System (edu track)

sprint planning

Sprint 1: Taskboard

1. user Registration and Login As
   Assigned to : Samyuktha

Iteration 1 : Taskboard

2. add and view student Records
   Assigned to Pavithra

Iteration 2: Taskboard

3. Update and Delete student Records
   Assigned to : Parkavi

Iteration 3: Taskboard

4. CSV .File Upload for Bulk Import
   Assigned to : Samyuktha / Pavithra

5. Student Search Functionality
   Assigned to : Parkavi

Result: The sprints are created for the Student Data Management System (edu track)

EX NO:5

Poker Estimation

AIM:

To create poker Estimation for the user Stories Student data Management System (Edu track)

POKER Estimation:

Story Points measure :

* Effort
* complexity
* Time

This uses Fibonacci Scale : 1, 2, 3, 5, 8, 13 . . .

Story points for Student Data Management system

~~Story~~ User ~~Story~~	Story Points	Reasons
user Registration	3	Form handling + data storage.
Student Record Management (Add / View)	5	CRUD Operations + database interaction
update and Delete Student Records	5	Data modification + validation
CSV File upload	8	File handling + validation + bulk processing
Student Search Functionality	3	Query + filtering logic

Total = 24 Story points

Boards → Work items → User story →
Story points (1, 2, 3, 5, 8) → Save
Repeat for all user stories

Result:

The estimation / story points are created for the project using Poker Estimation.

EXP No 6

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

AIM:

To Design a class diagram and seauence Diagram for the student database system.

CLASS DIAGRAM:

* A class diagram is a type of UML diagram that represents the structure of a system by showing classes. Attributs (variables). methods (functions). ~~classes~~ , ~~attributes~~ Relationships (association. inheritance . etc...

* It is used to design and understand architecture.

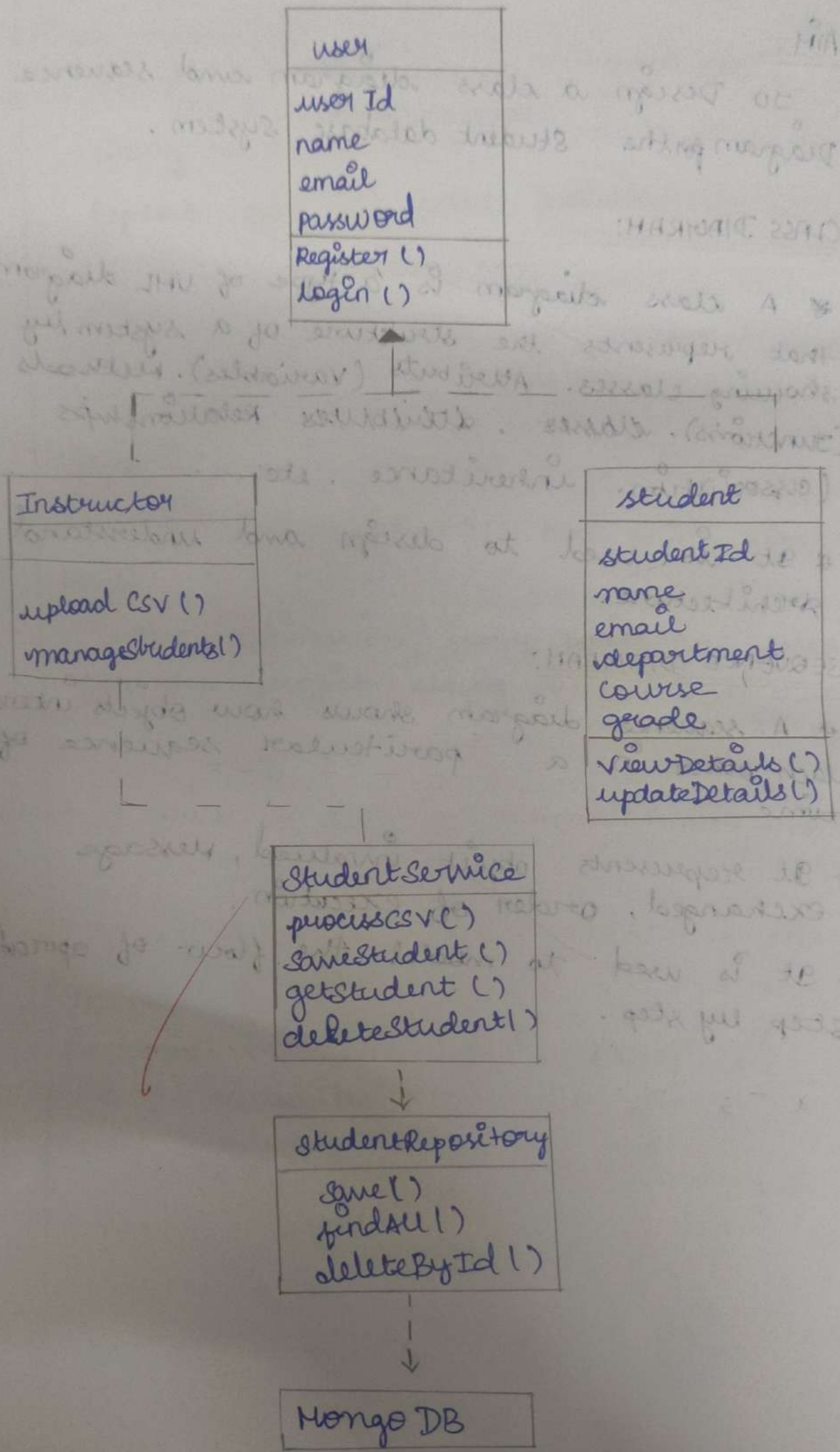
SEQUENCE DIAGRAM:

* A seauence diagram shows how objects interact each other in a paritcular seauence of time.

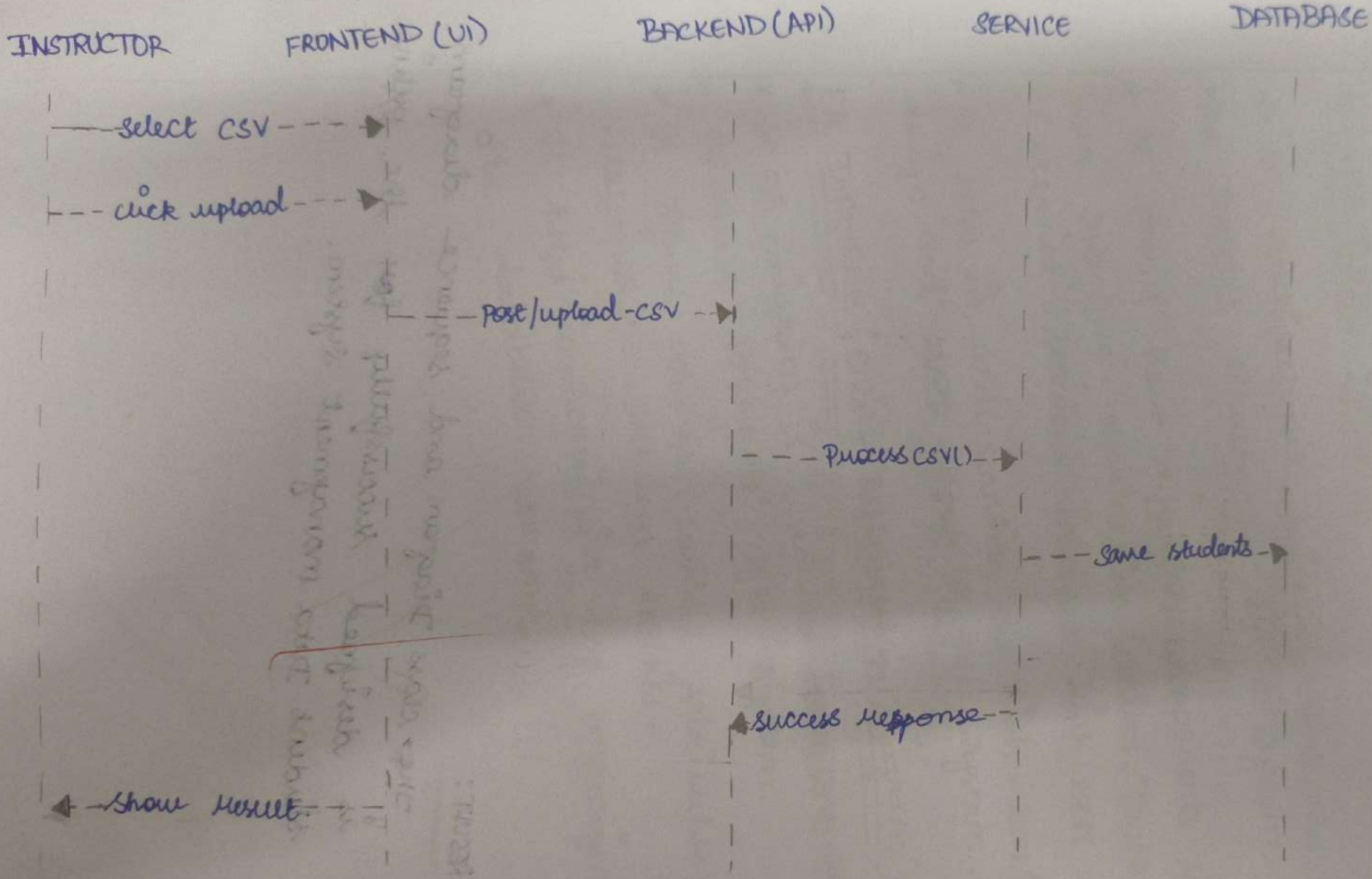
* It represents object invalued, message exchanged, order of execution.

* It is used to model the flow of operation step by step.

CLASS DIAGRAM :



SEQUENCE DIAGRAM :-



RESULT :

The class Diagram and sequence diagram is designed successfully for the ~~online~~ student Data management system.

NO:7 DESIGNING ARCHITECTURE DIAGRAM AND ER DIAGRAM

AIM:

To Design an architectural diagram and ER diagrams for the given project.

ARCHITECTURAL DIAGRAM:

* An architectural diagram represents the the overall structure of a system

* It shows how different components like user interface. Application / server. Database External services interact with each other.

* It helps in understanding the high-level design and data flow of the system.

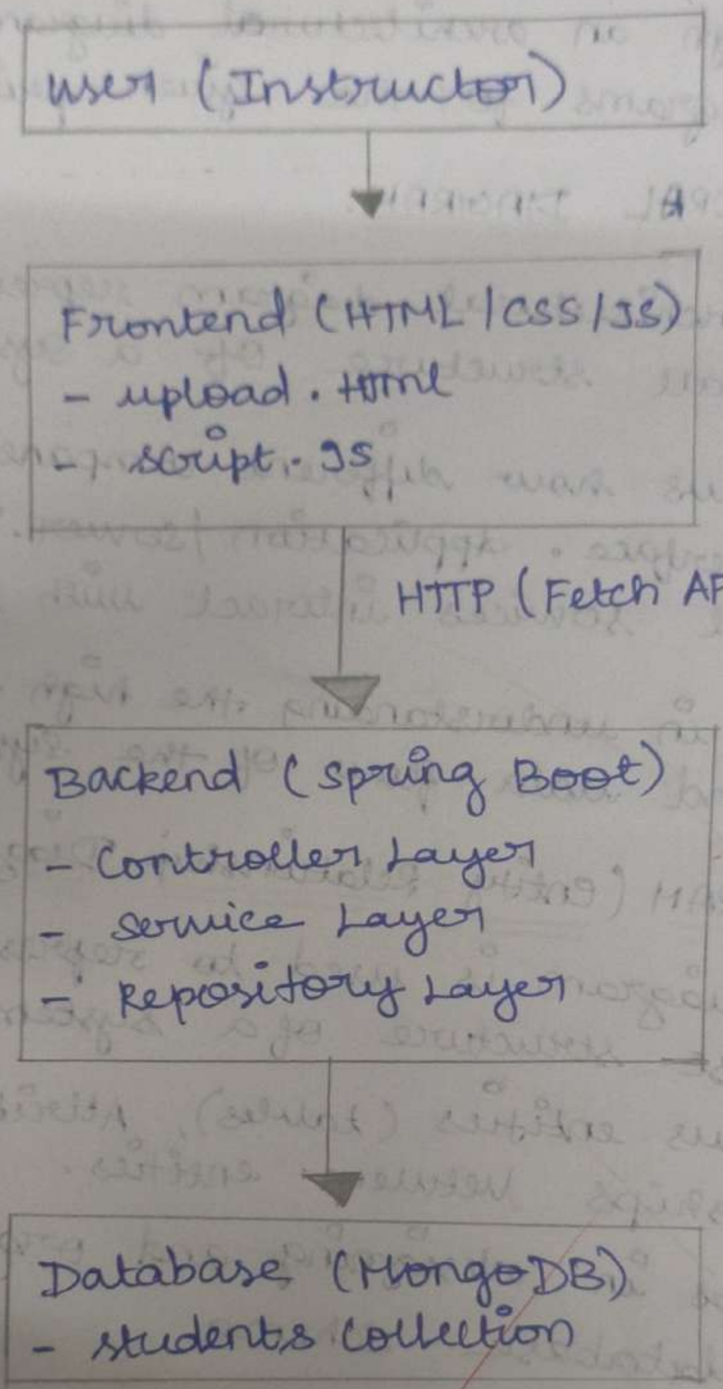
ER DIAGRAM (Entity Relationship Diagram)

* An ER diagram is used to represent the database structure of a system.

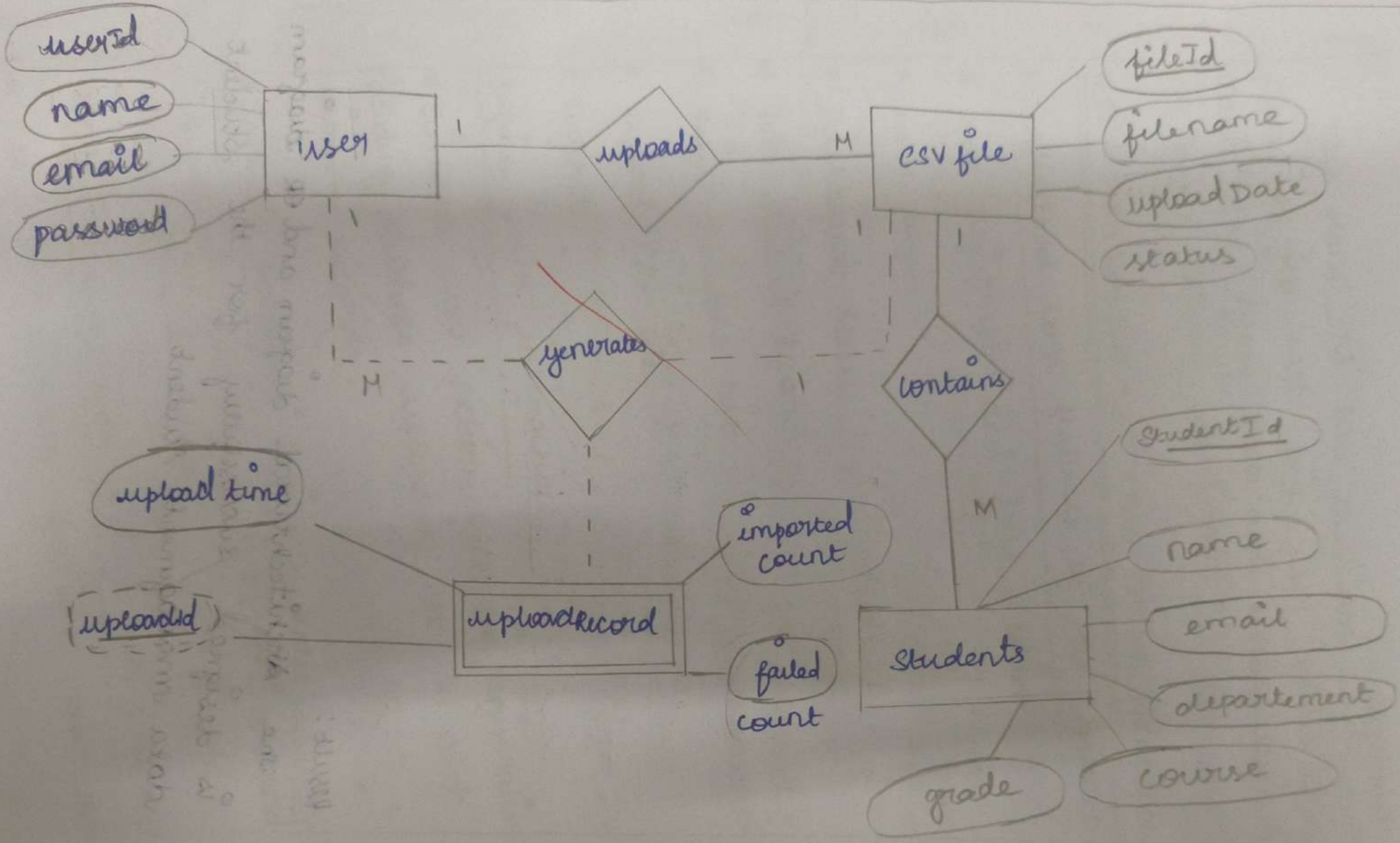
* It shows entities (tables), Attributes (Fields) Relationships between entities.

* It helps in designing and organizing the database schema.

ARCHITECTURE DIAGRAM



ER DIAGRAM


userId
name
email
password
user
1
uploads
M
csv file
fileId
filename
uploadDate
status
1
1
1
generates
M
1
contains
M
StudentId
name
email
departement
course
grade
Students
upload time
uploadid
uploadRecord
imported count
failed count

Result:

The architechtural diagram and ER diagram is designed successfully for the student data management student.

EXP NO 8

TESTING - TEST PLANS AND TEST CASES

AIM:

Test plans and test case and write two test cases for at least five user stories showing the happy path and errors scenarios in Azure Devops platform.

TEST PLANING AND TEST CASE DESIGN PROCEDURE:

1. understand core features of the Application:

* user registration and ~~student~~ login
* user login
* Dashboard access
* Add student Record
* view student list
* upload CSV file
* Logout.

Define user interaction

* simulate user action such as registration, login adding students, uploading csv files and viewing records.
* Include incorrect login credentials, invalid csv upload, missing student details, and unauthorized access.

3. Design happy path and error path test cases

* Focus on validating features under normal conditions

Example: Admin uploads Valid CSV and student data is stored successfully.

4. Design error path test cases

* simulate negative scenarios to test robustness

* Example: Login fails with incorrect password invalid file type upload, empty CSV file, duplicate student ID.

5. Breakdown steps and expected results.

* each test case includes clear actions and expected outcomes.

* Ensures clarity for both manual and automat testing

6. user clear naming and IDs

* TC01 - successful Registration
* TC02 - Invalid Login.
* TC03 - CSV upload success
* TC04 - Invalid CSV upload.

7. Separate test suites

* Authentication
* student Management
* CSV upload
* Dashboard

8. priorityze and review

Critical actions like login, registration, and CSV upload are marked as high priority.

TEST SUITE TS01 - user Authentication.

TC01 - Successful Registration

Action:

* Open register page
* Enter valid username, email, password
* click "Register"

Expected Result:

* Account created successfully
* user redirected to login page

Type:

Happy path

TC02 - Registration with Existing

Action:

* open register page
* enter already registered email
* click "Register"

Expected Result:

* Error message: "Email already exists"

Type:

Error Path.

TC03 - Successful Login

Action:

* open login page
* Enter valid email and password
* click "Login"

Expected Result:

* user redirected to dashboard

Type:

Happy path

TC04 - Login with Invalid Password

Action:

* open Login page
* Enter valid email + wrong password
* click "login"

Expected Result:

* Error message: "Invalid credentials"

Type:

Error Path

TEST SUITE TS02 - student Management

TC05 - Add student successfully

Action:

* Navigate to Add student page
* Enter valid student details
* click "save"

Expected Result:

* student added successfully
* student appears in student list

Type:

Happy path.

TC06 - Add student with Missing Fields

Action:

* Leave mandatory fields blank
* click "save"

Expected Result:

* validation message shown
* Record not saved

TYPE:

ERROR Path.

TEST SUITE TS03 - CSV UPLOAD

TC07 - upload valid CSV file

Action:

* Navigate to upload csv
* select valid csv file
* click "upload"

Expected result:

* File uploaded successfully
* student records imported into database

Type:

Happy path

TC08 – upload Invalid File Format

Action:

* Select non-csv file
* click "upload"

Expected Result:

* Error message: "only csv file allowed"

Type:

Error path

TC09 – upload Empty csv

Action:

* select empty CSV file
* click "upload"

Expected Result:

* error message displayed
* no records imported

Type: Error Path

TEST SUITE TS04 – DASHBOARD

TC10 – View Dashboard successfully

Action:

* Login successfully
* open dashboard

Expected Result:

* Dashboard loads with student statistics

Type: Happy path

TC11 - UNAUTHORIZED DASHBOARD ACCESS

Action:

* open dashboard without login

Expected Result:

* Redirect to login page

Type

security / Error Path

TEST SUITE TS05 - Logout

TC12 - Successful logout

Action:

* click logout button

Expected Result:

* session cleared
* Redirected to login page

Type:

Happy path.

USER STORIES

* As an admin, I want to register securely
* As an admin, I want to log in and access my dashboard
* As an admin, I want to add student records manually.
* As an admin, I want to upload csv files for bulk student records.
* As an admin, I want to view all uploaded student data.

Result:

The test plans and test cases for the user stories is created in Azure Devops with Happy path and error path.

EXPNO9 LOAD TESTING AND CI/CD PIPELINE

AIM:

To perform load testing on the Edu track student Management system web Application to evaluate its performance under multiple users, and to implement a CI/CD pipeline for automating the build, testing, and deployment process of the application.

LOAD TESTING:

* The load testing was performed on the Edu track / student Management System developed Spring Boot, MongoDB, HTML, CSS, and Javascript
* Multiple virtual users were simulated to access the application simultaneously including:
    * user Registration
    * user login
    * student Record Management
    * CSV upload
    * Dashboard Access

PIPELINE SETUP:

* A CI/CD pipeline was configured using Azure Dev ops by connecting the github repository containing the Edu track project.
* The pipeline automates:
    * Code checkout from github
    * Java / Maven environment setup
    * Dependency installation
    * Build process
    * Application execution / testing

YAML

```
trigger:
- main

pool:
  vmImage: 'ubuntu - latest'

steps:
- checkout : self
- task: JavaToolInstaller@0
  inputs:
    versionSpec: '17'
    jdkArchitectureOption: 'x64'
    jdkSourceOption : 'preInstalled'
  displayName: 'set up Java'
- script: |
    mvn clean install
  displayName: 'Install dependencies and build project'
- script: |
    mvn spring-boot:run
  displayName: 'Run Spring Boot Application'
```

RESULT:

The Student Management system successfully handle multiple user requests during load testing with stable performance, The CI/CD pipeline was successfully implemented, automating project build, dependency installation, testing, and execution ensuring continuous integration and smoother deployment.

EXP:10 GITHUB PROJECT STRUCTURE AND NAMING CONVENTIONS

AIM:

To organize the Edu track Student Management System project using a proper folder structure and naming conventions in github, making the project easy to understand maintain, and extend.

GITHUB USAGE:

The project was uploaded to a github repository for version control and collaboration. The repository includes all source code, frontend files, backend files, configuration files and doc, allowing easy access, updates, and project management.

PROCEDURE:

1. created the edu track Student Management System project using spring Boot, MongoDB HTML, CSS and JavaScript.

2. organized files into proper folders such as:

* frontend/ for UI pages
* src/main/java/com/edutrack/ for backend source code
* controller/ for request handling
* service/ for business logic
* repository/ for db operations
* model/ for data models
* resources/ for configuration files.

3. Followed proper naming conventions for
* Java classes (student.Java, user.Java)
* controllers (studentController.Java)
* services (studentService.Java)
* Repository interfaces
* HTML pages (register.html, login.html, upload.html)

4. Created a github repository for the EduTrack project
5. Uploaded all project files and maintained structured commits for version tracking
6. verified the project structure, readability and maintainability in the repository

RESULT:

The EduTrack Student Management system project was successfully organized using a clear folder structure and proper naming conventions. The github repository clearly displays the project architecture, making it easy for developers and users to understand maintain, and navigate the codebase efficiently.